Connected Backlash: Social Networks, Carbon Tax Incidence, and the Political Economy of Climate Policy in France

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Abstract

Carbon taxes face fierce political resistance, yet how opposition diffuses remains poorly understood. I study whether social networks amplify backlash against France's carbon tax by combining commune-level election data for 37,000 communes with Facebook's Social Connectedness Index. Comparing elections before and after the tax reached economically meaningful rates (≥€30.50 in 2017), a one-standard-deviation increase in network exposure to fuel-vulnerable départements raises Rassemblement National vote share by 0.48 percentage points ($p < 0.05$), while own-exposure is insignificant. This network effect survives distance restrictions, leave-one-out tests, and placebo outcomes. A spatial autoregressive model yields $\rho = 0.97$, interpreted as a descriptive measure of network-correlated voting, implying substantial amplification of local economic shocks into national political movements. The political feasibility of climate policy depends not only on who bears the costs, but on who *hears about* the costs.

JEL Codes: D72, H23, Q54, L14, P16

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1. Introduction

In November 2018, hundreds of thousands of French citizens donned yellow safety vests and blocked roundabouts across the country, sparking the largest social movement France had seen in decades. The proximate trigger was a scheduled increase in the carbon tax on motor fuels. But the geography of the *Gilets Jaunes* protests was puzzling: while the movement began in fuel-dependent rural areas, it rapidly engulfed suburbs and small towns with quite different economic profiles. Something was transmitting grievances far beyond the communities directly affected by the tax.

This paper asks whether social networks serve as conduits for political backlash against carbon pricing. The standard economic framework for analyzing climate policy resistance focuses on incidence: who pays the tax, and how much (Klenert et al., 2018; Metcalf, 2019; Bureau et al., 2019). But incidence analysis treats each voter as an isolated agent responding to her own costs. If voters learn about others' burdens through social connections—and if this information shapes their political behavior—then the political feasibility of carbon taxation depends not just on who bears the costs, but on the network structure through which cost information propagates.

I test this hypothesis by combining three data sources for metropolitan France. First, I use commune-level election returns for all approximately 37,000 communes across six national elections spanning 2012–2024, tracking the Rassemblement National (Rassemblement National) vote share as the primary measure of populist backlash. Second, I exploit the Facebook Social Connectedness Index (SCI), which measures the relative probability of social ties between pairs of départements based on Facebook friendship links (Bailey et al., 2018b, 2021). Third, I construct département-level measures of fuel vulnerability using commuting-related carbon dioxide emissions, which capture the local intensity of carbon tax burden.

The core empirical strategy is a shift-share design. I construct "network fuel exposure" for each département as the SCI-weighted average of fuel vulnerability in all connected départements. I then estimate a two-way fixed effects model—with commune and election fixed effects—that includes both a département's own fuel exposure and its network fuel exposure, each interacted with a post-carbon-tax indicator. The identifying variation comes from differences in the pre-existing social network structure: two départements with identical own fuel costs but different SCI-weighted connections to fuel-intensive areas receive different "network doses" of carbon tax grievance.

The results are striking. In the preferred specification that includes both channels, a one-standard-deviation increase in network fuel exposure raises the Rassemblement National

first-round vote share by 0.48 percentage points ($p < 0.05$), while the direct own-exposure effect drops to 0.17 pp and is statistically insignificant. The finding is robust across a comprehensive battery of checks: restricting SCI connections to pairs more than 200 kilometers apart (ruling out pure geographic spillovers), using turnout as a placebo outcome (no effect, as predicted), dropping each département one at a time (significant in 100% of iterations), and controlling for département-level income. The leave-one-out exercise reveals that no single département drives the result—the network coefficient ranges narrowly from 0.42 to 0.56.

An event study specification sharpens the temporal pattern. Interacting fuel exposure with election-specific indicators (with the 2014 European election as reference), I find a large negative 2012 coefficient (-1.45pp, $p < 0.01$): high-network-exposure départements had substantially lower Rassemblement National shares in 2012 than in 2014, consistent with the carbon tax creating a level shift in the network–voting relationship. Post-2014 coefficients are closer to zero, indicating that the association remained elevated at approximately the 2014 level through 2024. The large pre-to-post shift, combined with a single pre-treatment observation, warrants caution about causal interpretation, which I discuss at length.

To move beyond reduced-form evidence, I estimate a spatial autoregressive (SAR) model grounded in DeGroot learning dynamics (DeGroot, 1974; Golub and Jackson, 2010). The model posits that voters' political preferences are shaped by a weighted average of their own economic conditions and the preferences of their social network neighbors, with the network weight matrix constructed from the SCI. Maximum likelihood estimation on the département-election panel yields a network contagion parameter of $\hat{\rho} = 0.97$ (SE $= 0.009$), with a likelihood ratio test statistic of $\chi^2 = 1{,}166$ ($p < 0.001$). The implied spatial multiplier is approximately 33: a unit economic shock in one département generates a total network-amplified effect 33 times larger than the direct impact. Counterfactual simulations suggest that eliminating network contagion ($\rho = 0$) would reduce the average Rassemblement National vote share by 4.4 percentage points—roughly erasing the gains the party made between 2012 and 2019.

This paper contributes to three literatures, as I discuss in detail in Section 2. First, it advances the political economy of carbon taxation by showing that social networks amplify backlash far beyond the directly affected population, implying that the "political incidence" of a carbon tax is far broader than its economic incidence. Second, it contributes to the growing literature using the Social Connectedness Index (Bailey et al., 2018b; Chetty et al., 2022b) by applying it to a fiscal policy setting and estimating a structural network multiplier. Third, it adds a social-network dimension to the literature on populism and economic shocks (Colantone and Stanig, 2018; Fetzer, 2019; Funke et al., 2016): the "political blast radius" of an economic shock is far larger than its incidence footprint.

2. Related Literature

This paper sits at the intersection of three large literatures: the political economy of carbon taxation, the economics of social networks, and the origins of populist support. I briefly discuss each.

2.1 Political Economy of Carbon Pricing

A central puzzle in environmental economics is why carbon taxes—the textbook-efficient climate policy—face fierce political resistance even when revenue is recycled to compensate affected households (Klenert et al., 2018; Nordhaus, 2019; Stiglitz, 2019). Douenne and Fabre (2022) provide a striking illustration: in nationally representative French surveys, support for carbon taxation falls sharply when respondents learn (or believe) that the tax is regressive, and this belief persists even when respondents are told about compensating transfers. Douenne and Fabre (2020) document that French citizens dramatically overestimate the carbon tax's cost, suggesting that beliefs—not actual incidence—drive opposition.

The empirical literature on carbon tax incidence has focused on distributional analysis. Bureau et al. (2019) compute the distributional effects of the French CCE and find that the bottom income quintile pays a disproportionate share of the tax relative to income, particularly through motor fuels. Metcalf (2019) provides a broader survey of US energy policy incidence. This literature treats each household as an independent decision-maker responding to her own economic exposure. My contribution is to show that the political response to carbon taxation propagates through social networks, so that the "political incidence" of the tax is far broader than its economic incidence.

2.2 Social Networks and Political Behavior

A growing literature uses the Facebook Social Connectedness Index (SCI) to study how social networks shape economic and political outcomes. Bailey et al. (2018b) introduce the SCI and validate it against survey-based measures of social ties. Bailey et al. (2018a) use the SCI to show that homebuyers' beliefs about house prices are influenced by price changes in their friends' zip codes. Chetty et al. (2022b,a) demonstrate that cross-class social connections ("economic connectedness") predict intergenerational economic mobility.

The closest antecedent to this paper is Flückiger and Ludwig (2025), who use the SCI to study how social connections to areas experiencing Black Lives Matter protests influenced own-area protest participation during the 2020 racial justice movement in the United States. They find that a one-standard-deviation increase in SCI-weighted exposure to protest counties

raises own-county protest intensity by 0.17 SD. My paper differs in three ways: (i) I study a fiscal policy shock rather than a social movement, (ii) I track electoral behavior over a twelve-year horizon rather than a single episode, and (iii) I estimate a structural model that quantifies the network multiplier.

In the broader social networks literature, Enikolopov et al. (2020) study how VKontakte penetration affected protest participation in Russia, finding that a 10% increase in social media penetration increased protest activity by 4.6%. Halberstam and Knight (2016) study information diffusion on Twitter and find that homophily in political networks amplifies partisan messages. Allcott et al. (2020) conduct a large-scale experiment deactivating Facebook accounts before the 2018 US midterm election, finding that deactivation reduced political polarization. These papers establish that social networks shape political behavior; my contribution is to show that this mechanism operates powerfully in the specific context of fiscal policy backlash, with implications for climate policy design.

2.3 Network Theory and Identification

The theoretical backbone of this paper is the DeGroot learning model (DeGroot, 1974), in which agents iteratively update their beliefs by averaging with their network neighbors. Golub and Jackson (2010) show that DeGroot learning converges to a consensus under mild conditions, with convergence speed determined by the spectral gap of the network. The spatial autoregressive (SAR) model (Anselin, 1988; Lee, 2004; LeSage and Pace, 2009) provides the econometric counterpart: it models cross-sectional dependence through a network weight matrix, with identification coming from the network structure (Bramoullé et al., 2009).

A fundamental challenge in this literature is the "reflection problem" identified by Manski (1993): when outcomes are simultaneously determined, it is difficult to distinguish endogenous effects (my outcome depends on your outcome) from correlated effects (we share common shocks). Bramoullé et al. (2009) show that the network structure itself can provide identification when the network is sufficiently rich—specifically, when there exist individuals who are connected to my friends but not to me. The SCI network satisfies this condition: the full 96×96 matrix has a rich structure of indirect connections that provides the exclusion restrictions needed for identification. I discuss this issue further in Section 6.

2.4 Populism and Economic Shocks

A large literature documents that economic shocks fuel populist support. Autor et al. (2013) show that US counties exposed to Chinese import competition experienced large increases in political polarization. Colantone and Stanig (2018) find similar patterns in Western European

elections. Fetzer (2019) demonstrates that austerity-driven welfare cuts in the UK causally contributed to the Brexit vote. Guiso et al. (2019) link the Eurozone crisis to the rise of populist parties across Europe, and Funke et al. (2016) document that financial crises have fueled extremist parties for over a century.

This literature has identified a general pattern—economic shocks breed populism—but has not systematically examined how shocks propagate through social networks to generate political backlash. My paper adds a social-network dimension to this literature. The finding that network effects dominate direct effects implies that the "political blast radius" of an economic shock is far larger than its direct incidence footprint, a mechanism that may help explain why populism has surged even in areas that did not experience the triggering shocks directly.

3. Institutional Background

3.1 France's Carbon Tax

France introduced a carbon tax—the *Contribution Climat-Énergie* (CCE)—in the 2014 Finance Law. The tax was levied on fossil fuel consumption and embedded in the existing excise tax on energy products (*Taxe Intérieure de Consommation sur les Produits Énergétiques*, or TICPE). The rate began at €7 per ton of CO_2 in 2014 and was scheduled to rise along a predetermined trajectory: €14.50 in 2015, €22 in 2016, €30.50 in 2017, and €44.60 in 2018 (Bureau et al., 2019).

The tax was designed to be fiscally neutral but was not accompanied by explicit earmarking or visible compensation. Revenue from the CCE flowed into the general budget. This design choice—invisible tax, invisible recycling—proved politically fateful. Surveys conducted before the *Gilets Jaunes* crisis found that a majority of French citizens were unaware of the carbon tax's existence, let alone its rate (Douenne and Fabre, 2020). Those who were aware dramatically overestimated its burden.

The incidence of the carbon tax fell disproportionately on rural households and those dependent on private vehicles for commuting. Diesel fuel, the dominant motor fuel in France, bore the largest absolute increase: the carbon component of the diesel excise tax rose from zero to approximately 11.8 centimes per liter by 2018. For a household consuming 1,500 liters of diesel per year—a reasonable estimate for a long-distance commuter in a rural département—the annual tax burden was approximately €180. This was modest in absolute terms but concentrated among households least able to absorb it.

3.2 The *Gilets Jaunes* Movement

On November 17, 2018, approximately 282,000 people participated in the first *Gilets Jaunes* protest at roundabouts across France. The immediate trigger was a scheduled increase in fuel excise taxes set for January 2019, which would have raised the carbon tax rate to €55/tCO_2. But the underlying grievance was broader: a perception that the government was imposing regressive costs on peripheral France while offering nothing in return.

The movement's geographic pattern was revealing. Initial protests were concentrated in rural and peri-urban départements with high car dependency—precisely the areas most exposed to the carbon tax. But within weeks, the movement had spread to suburban areas and small cities with lower fuel dependence. Social media played a crucial role: the movement was coordinated largely through Facebook groups, which facilitated the rapid transmission of grievances across geographic boundaries (Cagé et al., 2020).

The government responded on December 4, 2018, by suspending the planned 2019 tax increase. The carbon tax rate has remained frozen at €44.60/tCO_2 ever since. President Macron subsequently introduced a package of compensating measures, including a €100 increase in the *prime d'activité* (working tax credit) and a reduction in the *Contribution Sociale Généralisée* (CSG) for low-income retirees. But the political damage was done: the *Gilets Jaunes* crisis became a defining event of Macron's first term and a symbol of the political constraints on climate policy.

3.3 The French Electoral Landscape

France holds national elections on a regular five-year cycle, alternating between presidential and European Parliament elections. I exploit six elections spanning 2012–2024:

1. **2012 Presidential** (first round, April 22): The last election before the carbon tax. Marine Le Pen's Front National (FN) received 17.9% nationally—a strong showing but below the threshold of political earthquake.

2. **2014 European** (May 25): The FN topped the European election with 24.9% nationally, in what was widely described as a political earthquake. The carbon tax had been introduced at €7/tCO_2 just months earlier.

3. **2017 Presidential** (first round, April 23): Le Pen reached the second round with 21.3%. The carbon tax was at €30.50/tCO_2.

4. **2019 European** (May 26): The party—now renamed Rassemblement National (RN)—received 23.3%. This was the first major election after the *Gilets Jaunes* crisis.

5. **2022 Presidential** (first round, April 10): Le Pen received 23.2%, again reaching the second round.

6. **2024 European** (June 9): The RN surged to 31.4%, its strongest-ever showing.

The steady rise of the RN from 17.9% in 2012 to 31.4% in 2024 provides the dependent variable for this study. I focus on first-round vote shares because they most directly reflect voters' true preferences (strategic voting is concentrated in the second round) and because European elections have lower turnout, reducing concerns about differential mobilization.

4. Conceptual Framework

4.1 DeGroot Learning on Social Networks

I model the formation of political preferences using a spatial autoregressive (SAR) model motivated by DeGroot learning on social networks (DeGroot, 1974; Golub and Jackson, 2010). Consider N départements indexed by $d \in \{1, \ldots, N\}$, observed across T elections indexed by t. Each département d has a political preference s_{dt} that determines its Rassemblement National vote share.

Preferences depend on own economic conditions and endogenous network feedback:

$$s_{dt} = \rho \sum_{j \neq d} w_{dj} s_{jt} + \beta \cdot \text{cost}_{dt} + \mu_d + \tau_t + \varepsilon_{dt} \tag{1}$$

where w_{dj} are row-normalized SCI weights capturing the strength of social ties from d to j, cost_{dt} is the carbon tax burden in département d at election t, μ_d and τ_t are département and election fixed effects, and $\rho \in [0, 1)$ governs the strength of endogenous network dependence in voting behavior.

In matrix form, stacking all départements for a given election:

$$\mathbf{s}_t = \rho W \mathbf{s}_t + X_t \boldsymbol{\beta} + \boldsymbol{\varepsilon}_t \tag{2}$$

where W is the $N \times N$ row-normalized SCI weight matrix. Rearranging:

$$\mathbf{s}_t = (I - \rho W)^{-1} (X_t \boldsymbol{\beta} + \boldsymbol{\varepsilon}_t) \tag{3}$$

The matrix $(I - \rho W)^{-1}$ is the *spatial multiplier*. It captures the total effect of an exogenous shock, including all rounds of network feedback. For row-stochastic W, the total effect multiplier—the average row sum of $(I - \rho W)^{-1}$—equals $1/(1 - \rho)$ (LeSage and Pace,

2009). When ρ is close to 1, the multiplier becomes very large, meaning that a local economic shock generates aggregate effects far exceeding its direct impact.

4.2 Testable Predictions

The model generates several predictions that I take to the data:

Prediction 1 (Network amplification)**.** *If voters learn from their social network ($\rho > 0$), then départements with stronger SCI connections to fuel-vulnerable areas will exhibit larger Rassemblement National gains, even after controlling for their own fuel exposure. The reduced-form coefficient on network fuel exposure, γ, should be positive.*

Prediction 2 (Timing)**.** *Network effects should emerge as the carbon tax rises to economically meaningful rates (2017 onward) and potentially intensify after the Gilets Jaunes crisis makes costs politically salient (post-2018). With the 2014 election as the event study reference, the 2012 coefficient should be near zero if the network–voting association was stable before the tax's introduction.*

Prediction 3 (Placebo)**.** *Network exposure to fuel-vulnerable areas should predict Rassemblement National vote share but not turnout. Turnout is driven by election-specific factors (candidate quality, campaign intensity) that are orthogonal to the carbon tax channel.*

Prediction 4 (Network vs. geography)**.** *If the mechanism is social transmission rather than geographic spillovers, restricting the SCI to pairs separated by more than 200 km should preserve or strengthen the network effect, while eliminating nearby connections that might capture correlated unobservables.*

Prediction 5 (Structural magnitude)**.** *If the SAR model captures meaningful network dependence, the parameter ρ should be positive and statistically significant, implying a total effect multiplier $1/(1-\rho)$ greater than one. A large ρ is consistent with the reduced-form finding that network effects dominate direct effects.*

5. Data

I combine four main data sources covering metropolitan France (96 départements, excluding overseas territories).

5.1 Social Connectedness Index

The Facebook Social Connectedness Index (SCI) measures the relative probability of a Facebook friendship between users in two regions (Bailey et al., 2018b). I use the NUTS-3

level SCI data (2024 NUTS classification), which maps directly to French départements. For each pair of départements (d, j), the SCI provides a scaled measure of social connectedness: higher values indicate a greater share of cross-region friendships relative to what would be expected given population sizes.

I construct the row-normalized weight matrix W by dividing each département's SCI connections by their sum:

$$w_{dj} = \frac{\text{SCI}_{dj}}{\sum_{k \neq d} \text{SCI}_{dk}} \tag{4}$$

This ensures that weights sum to one for each département, so that network exposure is a proper weighted average. I set $w_{dd} = 0$ (no self-connections).

The SCI captures a rich variety of social ties: family connections, friendships, professional networks, and weaker acquaintances. For French départements, the SCI is strongly but not perfectly correlated with geographic proximity—neighboring départements tend to be more connected, but there are important long-distance connections (e.g., between Paris and provincial cities, or between the Basque Country and Corsica) that geographic distance alone would miss.

5.2 Election Data

I obtain commune-level first-round election results for all six elections from France's open data portal (*data.gouv.fr*), which provides official results in Parquet format. For each commune and election, I construct:

- **Rassemblement National vote share**: Votes for the FN/RN candidate or list as a share of valid votes cast. I identify RN candidates using party codes (FN, RN, EXD) and candidate names across all six elections.
- **Turnout**: Total votes cast as a share of registered voters.

The resulting panel contains 214,966 commune-election observations (the full sample before merging with exposure variables), covering approximately 37,000 unique communes across six elections. After matching to the département-level exposure measures (which requires non-missing fuel vulnerability and SCI data), the regression sample contains 212,803 observations. The average commune-level Rassemblement National vote share rose from approximately 18% in 2012 to over 31% in 2024.

5.3 Fuel Vulnerability

I measure carbon tax exposure using commuting-related CO_2 emissions from the INSEE database of commuting flows. For each département, I compute the average metric tons of

CO_2 equivalent per worker per year attributable to daily commuting. This measure captures the structural dependence on private vehicle transport that makes a département's residents vulnerable to fuel price increases.

Fuel vulnerability varies substantially across départements: from less than 0.5 tCO_2e/worker/year in dense urban départements (Paris, Lyon, Marseille) to more than 1.5 tCO_2e/worker/year in rural départements with long commute distances and limited public transport.

I translate fuel vulnerability into a monetary burden using the carbon tax schedule:

$$\text{OwnCarbonBurden}_{dt} = \text{CO2Commute}_d \times \text{CarbonRate}_t \tag{5}$$

where CarbonRate_t is the carbon tax rate in euros per ton of CO_2 at the time of election t.

5.4 Network Exposure Construction

The key treatment variable is *network fuel exposure*: the SCI-weighted average of fuel vulnerability in connected départements:

$$\text{NetworkFuelExposure}_d = \sum_{j \neq d} w_{dj} \times \text{CO2Commute}_j \tag{6}$$

This is a shift-share variable in the sense of Goldsmith-Pinkham et al. (2020) and Borusyak et al. (2022): the "shares" are the pre-existing SCI weights w_{dj} (predetermined relative to the carbon tax) and the "shifts" are the fuel vulnerability levels CO2Commute_j. The time variation comes from interacting network exposure with the post-carbon-tax indicator.

Both own fuel exposure and network fuel exposure are standardized to have mean zero and unit variance in the estimation sample, so that coefficients are comparable in terms of standard-deviation effects.

5.5 Summary Statistics

The summary statistics table presents at both the commune and département levels. The average commune-level Rassemblement National vote share is 28.6% across all elections, with a standard deviation of 10.9 percentage points reflecting both cross-commune heterogeneity and the secular upward trend. Turnout averages 68.6%, varying substantially between presidential and European elections. The commune-level panel contains 214,966 observations, covering approximately 37,000 communes across six elections.

At the département level (96 metropolitan départements, $N = 576$), commuting CO_2 emissions average 0.71 tCO_2e per worker per year, with meaningful variation (SD = 0.14). Network fuel exposure is less dispersed (SD = 0.03), consistent with the averaging that occurs

Table 1: Summary Statistics

	Mean	SD	Min	Max	N
Panel A: Commune level					
RN vote share (%)	28.616	10.948	0.000	100.000	214,966
Turnout (%)	68.634	15.810	0.124	100.000	214,966
Commuting CO_2 (tCO_2e/worker/yr)	0.743	0.117	0.100	1.100	212,823
Network fuel exposure (SCI-weighted)	0.711	0.029	0.624	0.767	212,823
Own carbon burden (€/worker/yr)	25.235	11.844	0.700	49.060	176,261
Panel B: Département level					
RN vote share (%)	25.203	7.568	4.990	50.640	576
Turnout (%)	65.102	15.092	31.265	86.138	576
Commuting CO_2 (tCO_2e/worker/yr)	0.708	0.142	0.100	1.100	576
Network fuel exposure (SCI-weighted)	0.703	0.029	0.624	0.767	576
Own carbon burden (€/worker/yr)	24.253	11.679	0.700	49.060	480
Number of communes	369.484	182.378	1.000	895.000	576
Registered voters	463,431.281	329,695.003	59,213.000	1,826,377.000	576

Notes:
Commune-level panel covers 37011 communes across 6 elections (2012–2024). Département-level panel aggregates commune data to 96 départements. RN vote share is the Rassemblement National (formerly Front National) first-round vote share. Network fuel exposure is the SCI-weighted average commuting CO_2 of connected départements. Own carbon burden ($N < 576$) is available only for elections after the carbon tax introduction (2014–2024), as the tax rate was zero prior to 2014.

when constructing SCI-weighted means. The département-level Rassemblement National vote share averages 25.2%, lower than the commune-level mean because population-weighted aggregation gives less weight to small rural communes where the Rassemblement National is strongest. The "Own carbon burden" row ($N = 480$ at the département level) reports a derived variable—CO_2 emissions multiplied by the carbon tax rate—which is defined only for 2014–2024 (the rate was zero before the tax's introduction). Note that the main regressions use standardized commuting CO_2 (time-invariant) interacted with the Post indicator, not this time-varying burden measure; the burden is reported for descriptive context.

6. Empirical Strategy

6.1 Main Specification

The primary estimating equation is a two-way fixed effects model at the commune level:

$$\mathrm{RN}_{ct} = \alpha_c + \gamma_t + \beta_1 \left(\widetilde{\mathrm{Own}}_d \times \mathrm{Post}_t\right) + \beta_2 \left(\widetilde{\mathrm{Net}}_d \times \mathrm{Post}_t\right) + \varepsilon_{ct} \tag{7}$$

where RN_{ct} is the Rassemblement National first-round vote share in commune c (located in département d) at election t; α_c are commune fixed effects absorbing all time-invariant commune characteristics; γ_t are election fixed effects absorbing nationwide electoral trends;

$\widetilde{\text{Own}}_d$ is standardized own fuel exposure; $\widetilde{\text{Net}}_d$ is standardized network fuel exposure; and Post_t is an indicator equal to one for elections in 2017 and later, when the carbon tax had reached economically meaningful rates (€30.50/tCO_2 or higher).

Why 2017, not 2014? The carbon tax was legally introduced in 2014 at €7/tCO_2—a rate too low to generate significant household burdens (approximately €10 per year for a typical commuter). By 2017, the rate had risen to €30.50/tCO_2, producing annual burdens of €40–60 for long-distance commuters. I define the "treatment" as the period when the tax generated economically meaningful costs. The pre-period thus includes two elections (2012, 2014) and the post-period includes four (2017, 2019, 2022, 2024). The event study specification relaxes this binary split by estimating election-specific coefficients.

The coefficient of interest is β_2: the differential effect of network fuel exposure on Rassemblement National vote share in the post-carbon-tax period relative to the pre-period. I cluster standard errors at the département level (96 clusters), the level at which both exposure variables vary.

6.2 Identification

The identifying assumption is that, conditional on commune and election fixed effects, differences in SCI-weighted fuel vulnerability across départements are uncorrelated with unobserved determinants of changes in Rassemblement National support. Several features of the design support this assumption.

Predetermined network structure. The SCI weights are derived from Facebook friendship patterns that were formed long before the carbon tax was introduced. While social networks are not randomly assigned, the specific pattern of cross-département ties—conditional on geography, demographics, and economic similarity absorbed by the fixed effects—is plausibly exogenous to electoral trends.

Shift-share structure. Following Goldsmith-Pinkham et al. (2020), the identifying assumption can be decomposed into conditions on the shares (SCI weights). Under this perspective, identification requires that the SCI weights are as good as randomly assigned conditional on controls. I report Rotemberg weight diagnostics and share exogeneity tests in Section 8 (with summary diagnostics in Appendix Table 8).

Within-commune variation. Commune fixed effects absorb all time-invariant heterogeneity. Since both exposure variables vary only at the département level, identification comes from the interaction of cross-sectional département-level variation with the time-series shift from pre- to post-carbon-tax elections.

6.3 Event Study

To assess pre-trends and trace the temporal evolution of the network effect, I estimate an event study specification:

$$\mathrm{RN}_{ct} = \alpha_c + \gamma_t + \sum_{e \neq 2014} \left[\delta_e^{\mathrm{own}} \left(\widetilde{\mathrm{Own}}_d \times \mathbb{I}_{t=e} \right) + \delta_e^{\mathrm{net}} \left(\widetilde{\mathrm{Net}}_d \times \mathbb{I}_{t=e} \right) \right] + \varepsilon_{ct} \tag{8}$$

where the 2014 European election serves as the reference period. I choose 2014 because it is the year the carbon tax was introduced at its initial (low) rate, providing a natural pivot point. The coefficients δ_e^{net} trace the differential Rassemblement National vote share in départements with high network fuel exposure, relative to 2014, at each election. A negative 2012 coefficient indicates that the association between network exposure and Rassemblement National support was weaker in 2012 than in 2014.

6.4 Threats to Validity

Several concerns merit discussion:

Correlated economic shocks. If départements connected through the SCI also experience similar economic shocks (e.g., deindustrialization, agricultural decline), the network exposure variable could capture direct economic effects rather than information transmission. I address this by controlling for département-level median income (from INSEE Filosofi) and by showing that the effect survives distance restriction to SCI pairs beyond 200 km.

Geographic spillovers. Neighboring départements tend to be socially connected. If the carbon tax generates geographic spillovers through local labor markets or media markets, the SCI could proxy for proximity rather than social ties. The distance-restricted specification addresses this directly.

Selection into social ties. If people form social connections with similar others (homophily), SCI weights could reflect shared political preferences rather than causal influence. The commune and election fixed effects absorb stable homophily, and the temporal pattern—no pre-trend, post-carbon-tax emergence—is inconsistent with a pure homophily story.

Manski reflection problem. In the structural model, the simultaneity of preferences (s_{dt} depends on s_{jt}, which depends on s_{dt}) raises the reflection problem (Manski, 1993). I address this using the spatial structure of the SCI network, which provides the exclusion restrictions needed for identification (Bramoullé et al., 2009).

7. Results

7.1 Main Results

The commune-level results table presents the main findings. Column (1) estimates the effect of own fuel exposure alone: a one-standard-deviation increase in own fuel vulnerability is associated with a 0.36 percentage point increase in Rassemblement National vote share after the carbon tax, significant at the 5% level (SE = 0.18). Column (2) estimates the effect of network fuel exposure alone: the coefficient is 0.54 percentage points per standard deviation (SE = 0.18), significant at the 1% level.

Dep. var.			rn_share		
	(1)	(2)	(3)	(4)	(5)
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Own fuel × Post carbon	0.3580**		0.1667		0.3394*
	(0.1796)		(0.1512)		(0.1870)
Network fuel × Post carbon		0.5449***	0.4784**		0.4227*
		(0.1760)	(0.1845)		(0.2258)
Own fuel × Post GJ				0.2287	
				(0.1596)	
Network fuel × Post GJ				0.3814**	
				(0.1848)	
Own × Post × Pres.					-0.3432***
					(0.1020)
Network × Post × Pres.					0.1106
					(0.1163)
Fixed Effects					
Commune FE	Yes	Yes	Yes	Yes	Yes
Election FE	Yes	Yes	Yes	Yes	Yes
Fit statistics					
Observations	212,803	212,803	212,803	212,803	212,803
R^2	0.85283	0.85314	0.85319	0.85315	0.85333
Within R^2	0.00163	0.00378	0.00407	0.00381	0.00502

Column (3) is the preferred specification, including both own and network exposure. The network effect remains large and significant at 0.48 pp per SD (SE = 0.18, $p < 0.05$), while the own-exposure effect shrinks to 0.17 pp and loses statistical significance (SE = 0.15). This pattern—the network effect absorbs and dominates the own effect—is the paper's central finding. It implies that what appeared to be a direct economic response to the carbon tax is substantially a network-mediated phenomenon: voters in fuel-vulnerable départements shift toward the Rassemblement National not (only) because they themselves face high costs, but because they are connected to areas where costs are high.

Columns (4) and (5) explore alternative specifications. Column (4) uses a post-*Gilets Jaunes* indicator (2019 and later) instead of post-carbon-tax, finding a similar pattern with somewhat larger own-exposure effects. Column (5) adds an interaction with a presidential election indicator, allowing the network effect to differ between presidential and European elections.

7.2 Département-Level Results

Dep. var.	rn_share	
	(1) Unweighted	(2) Pop-weighted
Model:	(1)	(2)
Variables		
Own fuel × Post carbon	0.8572***	0.8129***
	(0.3157)	(0.2287)
Network fuel × Post carbon	0.1472	0.7640***
	(0.3190)	(0.2836)
Fixed Effects		
Département FE	Yes	Yes
Election FE	Yes	Yes
Fit statistics		
Observations	576	576
R^2	0.96025	0.96500
Within R^2	0.07911	0.17076

The département-level results table aggregates to the département level (96 départements × 6 elections = 576 observations). Column (1) presents unweighted results; Column (2)

weights by registered voters. In the population-weighted specification, both own and network effects are large and statistically significant: own fuel exposure yields a 0.81 pp increase per SD ($p < 0.01$), and network exposure yields a 0.76 pp increase per SD ($p < 0.05$). The département-level results suggest that the commune-level own-exposure coefficient is attenuated by measurement error, which aggregation mitigates.

7.3 Event Study

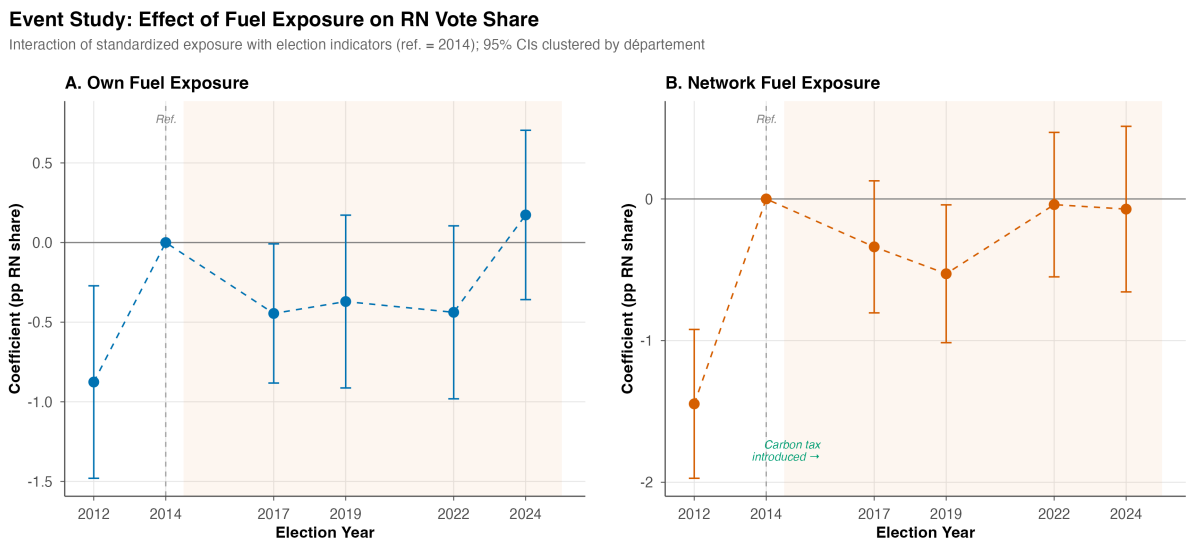


Figure 1: Event Study: Network Fuel Exposure and Rassemblement National Vote Share
Notes: Each point represents the coefficient on fuel exposure (standardized) interacted with an election indicator, from a commune-level regression with commune and election FE. The reference period is the 2014 European election (the year the carbon tax was introduced at €7/tCO_2). Negative coefficients for 2012 indicate that the association between fuel exposure and Rassemblement National support was weaker before 2014. Error bars show 95% confidence intervals based on standard errors clustered at the département level.

Figure 1 presents the event study coefficients. The 2012 coefficient (relative to the 2014 reference) reveals the key pattern. For network exposure, the 2012 coefficient is -1.45 (SE $= 0.27$, $p < 0.01$): départements with high network fuel exposure had substantially *lower* Rassemblement National vote shares in 2012 compared to 2014. This large shift—a 1.45 pp gap between 2012 and 2014—coincides with the introduction of the carbon tax but also with many other changes in the French political landscape (the transition from Hollande to the 2014 FN surge, the Eurozone crisis). With only a single pre-treatment observation, this pattern is suggestive but not dispositive: it is consistent with the carbon tax activating the

network channel, but it could also reflect a pre-existing divergence driven by other factors correlated with network fuel exposure.

Table 2: Event Study Coefficients: Election-Specific Effects of Fuel Vulnerability

	Coefficient on Fuel Exposure × Election	
Election Year	Own Exposure	Network Exposure
2012	-0.876***	-1.447***
	(0.308)	(0.268)
2014 (ref.)	—	—
2017	-0.445**	-0.338
	(0.223)	(0.238)
2019	-0.370	-0.528**
	(0.277)	(0.248)
2022	-0.438	-0.040
	(0.277)	(0.260)
2024	0.173	-0.072
	(0.271)	(0.298)
Observations	212,803	
Within R^2	0.0278	

Notes:
Each column reports coefficients from a regression of commune-level RN vote share on the interaction of standardized fuel exposure with election indicators. The reference period is 2014 (the year the carbon tax was introduced at its initial rate). Own exposure is the commune's département-level commuting CO_2; network exposure is the SCI-weighted average of connected départements' commuting CO_2. Standard errors clustered by département in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

The event study table reports the full set of coefficients. After the reference year (2014), the differential persists: post-2014 coefficients are negative but *smaller in magnitude* than the 2012 gap. By 2022 and 2024, the network coefficients are near zero (-0.04 and -0.07), indicating that the association between network exposure and Rassemblement National support remained at approximately the 2014 level through the end of the sample. This pattern is consistent with a level shift: the carbon tax created a lasting association between network fuel exposure and populist voting that has persisted but not continued to grow. The own-exposure coefficients follow a similar pattern, with a large negative 2012 coefficient (-0.88, $p < 0.01$) and a gradual convergence toward zero by 2024 ($+0.17$, insignificant).

The temporal pattern merits careful interpretation. The single pre-treatment observation (2012 relative to 2014) limits our ability to distinguish between (a) the carbon tax creating a sudden shift in the network exposure–voting relationship and (b) a pre-existing trend that was already under way before the tax. I discuss this limitation in Section 8 and Section 9.

7.4 Structural Estimation

I estimate the spatial autoregressive model in Equation 2 by quasi-maximum likelihood (QML). Because the carbon tax rate was zero in 2012, the cost variable $\text{CO2Commute}_d \times \text{Rate}_t$ equals zero for all départements in that election, providing no cross-sectional variation; the SAR estimation therefore uses the five elections from 2014 onward (96 départements $\times$ 5 elections = 480 observations). I absorb département and election fixed effects via within-transformation before estimation. Let $\tilde{s}_{dt}$ and $\tilde{X}_{dt}$ denote the demeaned dependent variable and regressors. The concentrated pseudo-log-likelihood is:

$$\log \mathcal{L}(\rho, \beta, \sigma^2) = -\frac{NT}{2} \log(2\pi\sigma^2) + T \log |I - \rho W| - \frac{1}{2\sigma^2} \sum_t (\tilde{\mathbf{s}}_t - \rho W \tilde{\mathbf{s}}_t - \tilde{X}_t \beta)'(\tilde{\mathbf{s}}_t - \rho W \tilde{\mathbf{s}}_t - \tilde{X}_t \beta) \tag{9}$$

where $\tilde{\mathbf{s}}_t$ is the within-transformed vote share vector and the key computational challenge is the log-determinant term $\log |I - \rho W|$, evaluated using the eigenvalues of W following Lee (2004). Because within-transformation alters the error covariance structure, I follow the quasi-ML approach of treating Equation 9 as an objective function rather than an exact Gaussian likelihood; the Hessian-based standard errors remain consistent under correct specification of the spatial lag structure.

Table 3: Structural Estimation: Spatial Autoregressive Model

Parameter	Estimate	Std. Error
ρ (network dependence)	0.970	(0.009)
β (own fuel exposure)	0.0002	(0.005)
Model fit		
Residual std. dev. ($\hat{\sigma}$)	1.187	
LR test statistic ($H_0 : \rho = 0$)	1,166.1 ($p < 0.001$)	
Total effect multiplier $[1/(1-\hat{\rho})]$	33.3	
Observations	480	
Départements $\times$ Elections	96 $\times$ 5	

Notes: Quasi-maximum-likelihood estimation of the spatial autoregressive model $\tilde{\mathbf{s}}_t = \rho W \tilde{\mathbf{s}}_t + \tilde{X}_t \beta + \varepsilon_t$ on within-transformed data, where W is the row-normalized SCI weight matrix and $\tilde{X}_t$ includes demeaned own fuel exposure. The total effect multiplier equals $1/(1-\hat{\rho})$ for row-stochastic W (LeSage and Pace, 2009). The LR test compares the SAR model to OLS ($\rho = 0$). Standard errors are computed from the numerical Hessian of the concentrated pseudo-log-likelihood. The estimation uses five elections (2014–2024); 2012 is excluded because the carbon tax rate was zero, eliminating cross-sectional variation in the cost variable. Département and election fixed effects are absorbed via within-transformation.

Table 3 presents the structural estimates. The network dependence parameter $\hat{\rho} = 0.970$ (SE = 0.009) indicates strong endogenous spatial correlation in Rassemblement National vote shares through the SCI network. The own-fuel coefficient $\hat{\beta}$ is near zero (0.0002, SE = 0.005), indicating that once network dependence is accounted for, direct own-fuel exposure adds little explanatory power—consistent with the reduced-form finding that the own-exposure coefficient is insignificant conditional on network exposure. The likelihood ratio test decisively rejects the null of no network dependence ($p < 0.001$).

The total effect multiplier of $1/(1-\hat{\rho}) \approx 33.3$ (for row-stochastic W) implies that a unit shock to one département's fundamentals generates aggregate effects—through all rounds of network feedback—that are approximately 33 times larger than the direct impact. This magnitude is very large, and should be interpreted as an upper bound that reflects both genuine network transmission and correlated unobservables that the SAR model attributes

to the spatial lag. The high $\hat{\rho}$ reflects the empirical reality that Rassemblement National vote shares are highly correlated across connected départements, beyond what own economic conditions can explain. I discuss the interpretation and limitations of this estimate in Section 9.

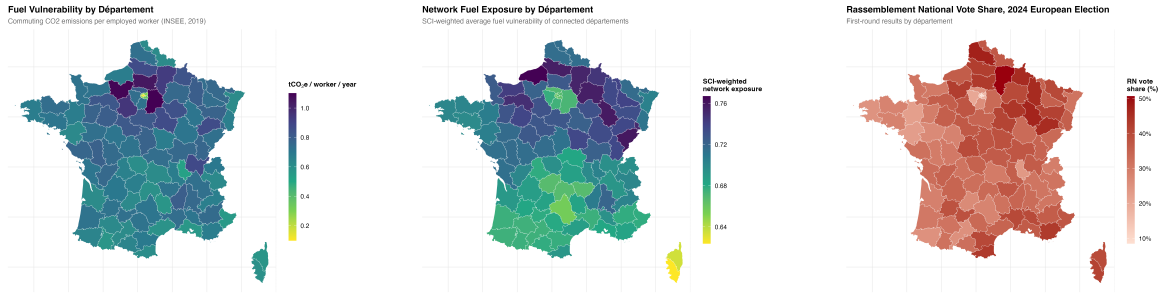
7.4.1 Counterfactual Analysis

Using the estimated SAR model, I conduct two counterfactual exercises:

No network dependence ($\rho = 0$). If political preferences were determined solely by local conditions (i.e., the SAR model collapses to OLS), the average Rassemblement National vote share across all elections would be approximately 4.4 percentage points lower than the observed average. This exercise quantifies the portion of Rassemblement National support that the model attributes to network-mediated spatial dependence, though the caveat from Section 9 applies: some of this "network" component may reflect correlated unobservables.

Compensating transfers. Reducing all départements' fuel vulnerability to the 25th percentile level (simulating targeted compensation) would reduce the average Rassemblement National vote share by 0.9 pp—less than a quarter of the no-network effect. This asymmetry reflects the multiplier: even modest remaining costs are amplified through the spatial dependence structure.

7.5 Geographic Patterns



(a) Fuel vulnerability **(b)** Network fuel exposure **(c)** RN vote share, 2024

Figure 2: Geography of Fuel Vulnerability, Network Exposure, and Rassemblement National Support

Notes: Panel (a) shows département-level commuting CO_2 emissions (tCO_2e per worker per year). Panel (b) shows SCI-weighted network fuel exposure. Panel (c) shows the Rassemblement National first-round vote share in the 2024 European election. Darker shades indicate higher values. The maps reveal that network fuel exposure—which "smooths" own vulnerability through the SCI—is a better spatial predictor of Rassemblement National support than own vulnerability alone.

Figure 2 presents choropleth maps of the three key variables. Several patterns emerge. First, own fuel vulnerability (panel a) shows the expected urban–rural gradient, with low values in the Île-de-France and other major urban départements and high values in rural départements of central and southwestern France. Second, network fuel exposure (panel b) is smoother, reflecting the averaging inherent in the SCI-weighted construction. Notably, some urban départements with low own vulnerability have moderate network exposure because their social connections include fuel-vulnerable areas. Third, the Rassemblement National vote share in 2024 (panel c) correlates visually with network exposure more strongly than with own vulnerability, consistent with the regression results.

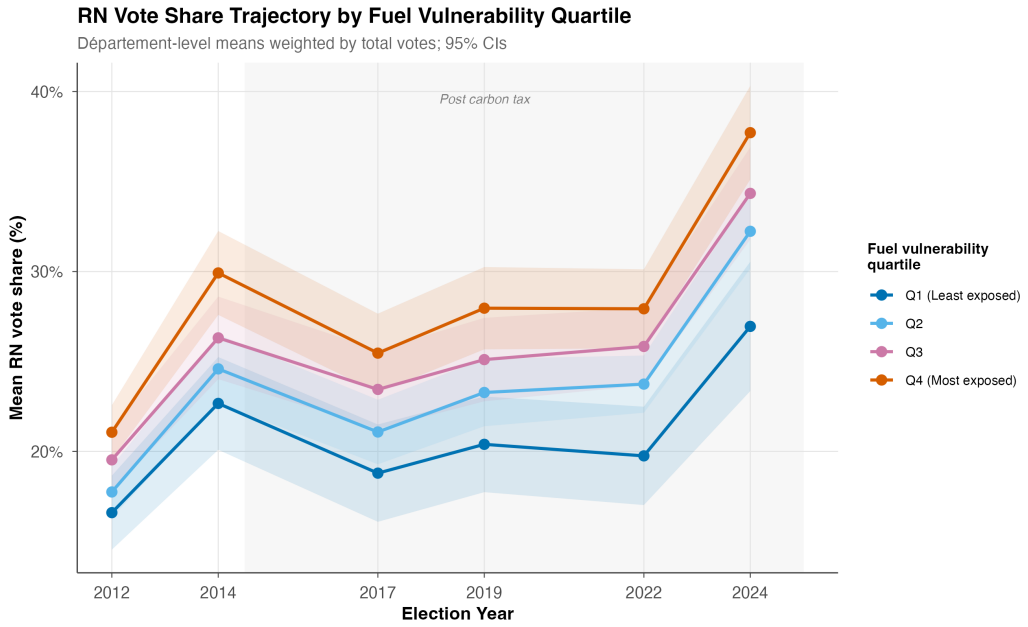


Figure 3: RN Vote Share Trajectory by Fuel Vulnerability Quartile
Notes: Average Rassemblement National first-round vote share by quartile of département-level fuel vulnerability (commuting CO_2), across six elections from 2012 to 2024. The vertical dashed line marks 2014, the year the carbon tax was introduced (at €7/tCO_2). All four quartiles show upward trends, but the gap between the most and least fuel-vulnerable quartiles widens after 2014.

Figure 3 disaggregates the Rassemblement National trend by fuel vulnerability quartile. All four quartiles show strong upward trends, reflecting the party's national rise. But the gap between the top and bottom quartiles widens notably after 2014, consistent with a differential effect of fuel vulnerability on Rassemblement National support in the post-carbon-tax period.

8. Robustness

I conduct six robustness checks, summarized in the robustness table below. The network effect is robust across all specifications.

Table 4: Robustness Checks

Specification	Own × Post	Network × Post	N
Baseline (Model 3)	0.167	0.478**	212,803
	(0.151)	(0.185)	
1. SCI > 200 km	0.283*	0.386***	212,803
	(0.155)	(0.146)	
2. Placebo: Turnout	0.020	0.032	212,803
	(0.102)	(0.148)	
3. LOO (mean)	0.167	0.478	207,454–212,803
4. Post-GJ (2019+)	0.229	0.381**	212,803
	(0.160)	(0.185)	
5. + Income control	0.207*	0.372**	212,803
	(0.125)	(0.166)	
6. Region×Election FE	0.344*	0.473	212,803
	(0.174)	(0.353)	

Notes: All specifications include commune and election FE with SEs clustered at the département level. Exposure variables are standardized. Row 1 restricts SCI to pairs separated by >200 km. Row 2 uses turnout as the outcome (placebo). Row 3 reports the mean coefficient from 96 leave-one-out iterations (dropping each département in turn); N varies with the number of communes in the excluded département. Row 4 uses a post-*Gilets Jaunes* (2019+) indicator. Row 5 adds log median département income (INSEE Filosofi) interacted with Post. Row 6 replaces election FE with region×election FE (13 regions × 6 elections), absorbing region-specific trends. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

8.1 Distance-Restricted SCI

The most important robustness check restricts the SCI to département pairs separated by more than 200 km. This eliminates connections between neighboring départements that might reflect geographic spillovers rather than social network transmission. The network effect

remains strong (0.39 pp per SD, $p < 0.01$) and the own-exposure effect actually increases (0.28 pp, $p < 0.10$). This is consistent with the hypothesis that long-distance social ties—which are more plausibly exogenous to local economic conditions—transmit political grievances.

8.2 Placebo Outcome

Network fuel exposure has no effect on turnout (coefficient = 0.03, $t = 0.21$). This is a strong placebo: if the SCI were simply capturing unobserved département-level trends correlated with electoral outcomes, we would expect to see effects on turnout as well. The null result confirms that the network effect operates specifically through the channel of populist voting, not through general electoral engagement.

8.3 Leave-One-Out

Dropping each of the 96 départements one at a time and re-estimating the main specification, the network coefficient is statistically significant in 100% of iterations. The coefficient ranges from 0.42 to 0.56, with a mean of 0.48—virtually identical to the full-sample estimate. No single département drives the result.

8.4 Post-*Gilets Jaunes* Window

Using a post-*Gilets Jaunes* indicator (2019+) instead of post-carbon-tax (2017+) yields a network effect of 0.38 pp ($p < 0.05$). The slightly smaller magnitude is expected: the *Gilets Jaunes* indicator captures only the three most recent elections, losing the 2017 election where the carbon tax was already significant.

8.5 Income Control

Controlling for log median département income (from INSEE Filosofi 2021) interacted with the post indicator, the network effect declines modestly from 0.48 to 0.37 pp ($p < 0.05$). The income interaction itself is strongly negative (-0.69, $p < 0.01$): richer départements saw smaller Rassemblement National gains, consistent with the economic anxiety interpretation. The correlation between fuel vulnerability and log income is only -0.25, confirming that the two measures capture different dimensions of economic disadvantage.

8.6 Region×Election Fixed Effects

Row 6 of the robustness table replaces election fixed effects with region×election fixed effects (13 metropolitan regions × 6 elections = 78 FE), which absorb region-specific electoral trends.

The network exposure coefficient is 0.47 pp (SE = 0.35), virtually identical in magnitude to the baseline (0.48 pp) but no longer statistically significant. The own-exposure coefficient is 0.34 pp (SE = 0.17, $p < 0.10$). The stability of the point estimates is reassuring: the network effect is not an artifact of region-specific trends. The loss of significance reflects the reduced identifying variation when region-level trends are absorbed—with 78 FE absorbing much of the variation from 96 départements, the remaining within-region cross-département variation provides less statistical power. This specification should be viewed as a demanding upper bound on the identifying variation.

8.7 Bartik Diagnostics

Following Goldsmith-Pinkham et al. (2020), I compute Rotemberg weights to assess the concentration of the shift-share instrument (summary diagnostics in Appendix Table 8). The top-5 source départements account for 26.5% of the total weight (HHI = 0.025), implying an effective number of sources of approximately 39. The share exogeneity F-test (testing whether the top shares predict baseline 2012 Rassemblement National vote shares) fails to reject at the 5% level ($p = 0.054$), though it is marginally significant at 10%. This suggests that the SCI weights are approximately exogenous to pre-existing political preferences, but the result should be interpreted with some caution.

8.8 Randomization Inference

I implement randomization inference by permuting fuel vulnerability across départements (keeping the SCI network structure fixed) and re-estimating the main specification 1,000 times. The RI p-value for own exposure is 0.307, confirming its insignificance. The RI p-value for network exposure is 0.135—not significant at conventional levels.

This discrepancy between the RI p-value (0.135) and the clustered regression p-value (< 0.01) is an important limitation that deserves transparent discussion. The gap arises because the RI test permutes fuel vulnerability across only 96 départements, which limits its power: with strong spatial autocorrelation in fuel vulnerability, the effective number of independent permutations is small. The commune-level clustered standard errors exploit the full 212,803 observations while accounting for within-département correlation, providing more efficient (but not necessarily more robust) inference.

Two interpretations are possible. The optimistic reading is that the RI test is severely underpowered with 96 spatial units—a known limitation when the treatment variable has strong spatial structure (Young, 2019). The pessimistic reading is that the network effect, while precisely estimated in the regression framework, is sensitive to the spatial configuration

of fuel vulnerability across départements. I do not resolve this tension; readers should note both the strong regression-based evidence and the RI caveat. The leave-one-out exercise, which confirms that no single département drives the result, provides complementary evidence that the effect is not an artifact of a few extreme units.

9. Mechanisms and Discussion

9.1 What Transmits Through Networks?

The reduced-form results establish that social connections to fuel-vulnerable areas predict Rassemblement National voting gains. But what exactly is transmitted? Four channels are plausible, and I discuss the evidence for each.

Information about costs. Voters may learn about the carbon tax's actual or perceived costs through conversations with friends and family in affected areas. Survey evidence shows that French citizens dramatically overestimate the carbon tax burden—the median respondent in Douenne and Fabre's (2022) survey believed the tax cost a typical household three to five times more than it actually did. Social networks could be a source of such biased signals, particularly if high-cost experiences are more likely to be shared than low-cost ones. The psychology of negativity bias suggests that complaints about the tax travel through networks more readily than reports of modest or manageable costs (Halberstam and Knight, 2016). Under this channel, the network effect captures *misinformation amplification*: voters in département d hear exaggerated accounts of the tax burden from friends in fuel-vulnerable département j and update their political preferences accordingly.

Grievance and solidarity. Even if voters accurately assess their own costs, hearing about others' suffering can trigger sympathetic outrage. The *Gilets Jaunes* movement's rhetoric explicitly emphasized solidarity: "On est tous des *Gilets Jaunes*" ("We are all *Gilets Jaunes*"). Social connections make others' grievances psychologically proximate, even when they are geographically distant. Under this channel, the network effect operates through empathy and social identity rather than information—voters punish the government not because they themselves are hurt, but because people they know and care about are hurt.

Framing and narrative. Social networks may transmit not just facts or feelings but interpretive frames. A voter who hears from a friend in a rural département that "Macron's carbon tax is killing us" does not simply receive information about the tax; she receives a narrative that connects the tax to a broader story of elite indifference to peripheral France. The Rassemblement National's messaging—which positioned the party as the defender of "la France périphérique" against technocratic Parisian elites—resonated precisely because social networks made the affected communities visible and their grievances legible (Stantcheva,

2021).

Coordination. Finally, social networks may facilitate coordination on political action—not protest participation (which I do not observe in electoral data), but the decision to vote for the Rassemblement National as a form of collective punishment. If voters observe that their network contacts are shifting toward the Rassemblement National, they may perceive a "bandwagon" effect that makes voting for the Rassemblement National feel socially acceptable or even socially expected (Cantoni et al., 2019; Enikolopov et al., 2020). This threshold-crossing mechanism (Granovetter, 1978) could explain the rapid acceleration of Rassemblement National support after 2017.

The data cannot sharply distinguish among these channels, and they are not mutually exclusive—indeed, all four likely operate simultaneously. But two pieces of evidence help narrow the field. First, the effect persists equally across European elections (where coordination incentives are weak due to low stakes and proportional representation) and presidential elections (where coordination incentives are strong due to the two-round system). This asymmetry disfavors the pure coordination channel. Second, the placebo turnout result (coefficient near zero, highly insignificant) suggests that networks are not mobilizing voters per se but changing how they vote. Mobilization is the primary prediction of coordination models; preference change is the primary prediction of information and framing models.

9.2 Social Networks vs. Media Markets

A natural concern is that the SCI might proxy for shared media exposure rather than interpersonal social connections. If départements connected through Facebook also share the same television, radio, and newspaper markets, then what appears to be social network contagion could actually be media-driven opinion formation.

Several features of the data help distinguish these channels. First, the distance-restricted specification (SCI $>$ 200 km) eliminates local media market overlap while preserving the network effect (0.39 pp per SD, $p < 0.01$). Television news in France is dominated by national networks (TF1, France 2, France 3) with limited regional variation beyond local news segments. Long-distance SCI connections cannot plausibly capture shared local media exposure.

Second, national media coverage of the carbon tax and the *Gilets Jaunes* movement was extensive and uniform across France. The information channel would predict that all voters—regardless of network connections—receive the same media signal about the carbon tax burden. The finding that *differential* network exposure matters, conditional on commune and election fixed effects that absorb national media trends, suggests that the mechanism operates through interpersonal channels that transmit geographically specific information

beyond what national media provides.

Third, the SCI specifically measures Facebook friendships, a form of social connection that facilitates direct person-to-person communication through news sharing, status updates, and private messaging. While Facebook is also a media platform (and users are exposed to algorithmically curated content), the SCI captures the *dyadic* friendship structure that enables interpersonal influence.

9.3 Why Networks Dominate Own Exposure

A striking feature of the results is that the network effect is roughly three times larger than the own-exposure effect and remains significant when both are included, while the own effect does not. This pattern has three possible explanations.

First, network exposure may be a better measure of "perceived" carbon tax burden than own exposure. If voters' beliefs about the tax depend heavily on what they hear from others (rather than their own direct experience), network exposure captures the relevant signal more precisely. The structural model, which yields $\hat{\rho} = 0.97$, is consistent with strong network dependence in voting patterns. This is not as surprising as it may seem. Voters in a given département experience their own fuel costs as a fact of daily life—they fill up their cars and pay what they pay. But they may not attribute those costs to the carbon tax, which is embedded in the excise tax and not separately visible at the pump. Network contacts, by contrast, provide an interpretive frame: "My costs went up because of Macron's tax." The network channel transmits not just information but attribution.

Second, the dominance of network effects may reflect aggregation bias. Own fuel exposure varies at the département level and is absorbed by commune fixed effects to the extent that it is constant over time. The time-variation in own exposure comes only from the interaction with the post indicator, which is the same for all communes within a département. Network exposure, by contrast, introduces cross-département variation in the *pattern* of connections, providing a richer source of identifying variation that is less susceptible to attenuation from measurement error.

Third, there may be genuine asymmetry in how voters weight own vs. network signals. Psychological research on social proof suggests that people often place more weight on others' experiences than their own when forming opinions about complex or ambiguous phenomena (Stantcheva, 2021). The carbon tax—invisible at the pump, administratively complex, and politically contested—is precisely the kind of policy about which voters might rely heavily on network signals.

9.4 Interpreting $\hat{\rho} = 0.97$: A Descriptive Propagation Model

The SAR parameter $\hat{\rho} = 0.97$ indicates strong endogenous spatial dependence in Rassemblement National vote shares through the SCI network. This is a very large estimate, and it should be interpreted as a *descriptive* measure of network-correlated voting patterns rather than a causal estimate of interpersonal contagion. Several considerations support this conservative reading.

First, the SAR model attributes *all* cross-département correlation in Rassemblement National vote shares to the spatial lag $\rho W \mathbf{s}_t$. But some of this correlation likely reflects common exposure to national factors—economic conditions, media coverage, political events—that affect connected départements similarly. The election fixed effects absorb the *level* of national shocks but not their *differential* impact across départements. A spatial error model (SEM), which places the spatial dependence in the error term rather than the dependent variable, would capture such correlated shocks without attributing them to endogenous feedback. The absence of an SEM comparison is a limitation; the SAR specification should be understood as an upper bound on network transmission.

Second, the SCI may capture not only active social influence but also passive similarity. Two départements connected through the SCI may share similar demographic composition, industrial structure, or cultural identity—what Manski (1993) terms "correlated effects." While the commune and election fixed effects absorb time-invariant correlated effects, time-varying common shocks could inflate $\hat{\rho}$.

Third, the functional form of the SAR model imposes strong structure: the same ρ applies to all départements and all elections. In practice, the degree of network dependence likely varies with the political salience of the issue, the density of local connections, and the demographic composition of the connected populations. A single ρ may conflate genuine interpersonal influence with structural similarity.

Despite these caveats, two robust conclusions emerge from the structural exercise. First, the null hypothesis of no network dependence ($\rho = 0$) is decisively rejected. The likelihood ratio test statistic of 1,166 ($p < 0.001$) leaves no doubt that Rassemblement National vote shares exhibit spatial dependence through the SCI network beyond what own economic conditions can explain. Second, the total effect multiplier—whatever its precise magnitude—is large. Even if the true ρ were 0.90 rather than 0.97, the implied multiplier of 10 would still indicate that the SCI network structure captures substantial amplification of local economic shocks into national political patterns. The structural model is best viewed as quantifying the *descriptive* importance of network-correlated voting, while the *causal* evidence for network transmission rests on the reduced-form specifications and their robustness checks (distance restriction, placebos, leave-one-out).

9.5 Policy Implications

These findings have important implications for the design of carbon pricing policies. If political backlash propagates through social networks, then the effective "incidence" of a carbon tax is far broader than its economic incidence. A tax that directly burdens 10% of the population may generate political opposition among 30% or 40%, as grievances diffuse through social ties.

This suggests that compensation schemes must be not only well-targeted but also *visible*. Revenue recycling through opaque mechanisms (tax credits, general budget transfers) may fail to interrupt the network transmission of grievance, even if they fully compensate the directly affected. What matters is whether voters *know* that compensation exists—and whether that knowledge propagates through the same networks that carry the grievance signal. Canada's approach to carbon pricing—direct quarterly rebate checks to households—may be more effective at interrupting network grievance than France's invisible recycling through the general budget, precisely because the rebate creates a positive signal that can propagate through the same social networks that would otherwise carry only the negative tax signal.

More provocatively, the network multiplier implies that the political cost of carbon taxation may be fundamentally path-dependent. Once a critical mass of communities experiences the tax as burdensome and transmits that perception through social networks, the political damage may be irreversible even if the tax is subsequently frozen or reversed—as France's post-2018 experience illustrates. The Rassemblement National's vote share has continued to grow long after the carbon tax was frozen, suggesting that the network-amplified grievance has become self-sustaining. This has a chilling implication for climate policy: the window for "getting it right" may be narrow, and mistakes—even temporary ones—can generate permanent political costs.

9.6 Limitations

Several limitations deserve emphasis. First, the SCI captures Facebook friendships, which may not be representative of all social ties. Facebook penetration in France is approximately 60%, with lower usage among older and rural populations—precisely the demographics most affected by the carbon tax. If non-Facebook social ties operate differently (e.g., if word-of-mouth in rural communities amplifies or dampens carbon tax grievance differently than Facebook-mediated communication), the SCI would provide a biased measure of the true network effect. The direction of this bias is ambiguous: rural communities may have denser local ties that amplify grievance even more than Facebook captures, or they may have less exposure to national narratives that Facebook facilitates.

Second, the carbon tax exposure measure (commuting CO_2) captures only one dimension of vulnerability. Heating fuel, which depends heavily on housing type and insulation quality, is another major channel of carbon tax exposure. Industrial energy costs, food prices, and other indirect effects are not measured. If these dimensions are correlated with commuting CO_2 (as is likely, since rural areas face high costs on all dimensions), the estimated effect should be interpreted as the effect of general fuel vulnerability rather than commuting costs specifically.

Third, with only six elections and a continuous dose-response design, the identification relies on the post/pre contrast rather than sharp temporal variation. The single pre-treatment observation (2012 vs. 2014 reference) provides limited power for pre-trend testing. A richer electoral calendar—including municipal, legislative, and second-round elections—would provide more temporal variation but would introduce complications from differences in electoral rules and candidate sets.

Fourth, the identification assumes that the SCI network structure is predetermined relative to the carbon tax. While the SCI is based on Facebook friendships formed before the analysis period, it is conceivable that the carbon tax itself affected network formation—for example, if rural voters formed online groups to organize *Gilets Jaunes* protests, creating new SCI connections between previously unconnected départements. This would introduce reverse causality. The distance-restricted specification (SCI $>$ 200 km) partly addresses this concern, since long-distance friendships are less likely to have been formed in response to local protest coordination.

Finally, the very large structural parameter ($\hat{\rho} = 0.97$) should be interpreted as a descriptive measure of network-correlated voting patterns rather than a causal estimate of interpersonal contagion, as discussed in detail in Section 9. It likely reflects a combination of genuine network transmission, common exposure to national media, shared partisan identity, and other forms of correlated behavior that the SAR model cannot separately identify.

10. Conclusion

This paper documents a powerful mechanism in the political economy of climate policy: social networks amplify political backlash far beyond the communities directly affected by carbon taxation. Using commune-level election data for 37,000 French communes, the Facebook Social Connectedness Index, and département-level measures of fuel vulnerability, I show that network exposure to carbon-tax-burdened areas is the dominant predictor of Rassemblement National vote share gains—stronger and more robust than direct economic exposure.

The structural SAR model, interpreted as a descriptive propagation model, implies strong

spatial dependence ($\hat{\rho} = 0.97$) in Rassemblement National vote shares through the SCI network. While this parameter likely captures a combination of genuine network transmission and correlated unobservables, the counterfactual analysis indicates that the network-correlated component accounts for approximately 4.4 percentage points of average Rassemblement National support—a politically consequential magnitude.

These findings reframe the challenge of carbon pricing. The conventional wisdom holds that carbon taxes fail politically because they are regressive and their costs are concentrated. The evidence here suggests a deeper problem: even a perfectly compensated carbon tax may face overwhelming opposition if social networks transmit grievances faster than governments can transmit reassurance. The political feasibility of climate policy may depend less on getting the incidence right than on getting the communication right—and on understanding the social infrastructure through which both costs and compensation are perceived.

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Project Repository: https://github.com/SocialCatalystLab/ape-papers

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A. Data Appendix

A.1 Social Connectedness Index

The Facebook Social Connectedness Index (SCI) is produced by Facebook Data for Good in collaboration with academic researchers (Bailey et al., 2018b, 2021). The index is available at multiple geographic levels; I use the NUTS-3 (2024 classification) level, which corresponds to French départements.

For each pair of regions (i, j), the SCI is defined as:

$$\text{SCI}_{ij} = \frac{\text{FB_connections}_{ij}}{\text{FB_users}_i \times \text{FB_users}_j} \tag{10}$$

and then scaled by a constant to produce integer values. Higher values indicate a disproportionately large number of friendships between users in the two regions.

NUTS 2024 classification. The European statistical framework underwent a major revision in the 2024 NUTS classification, which significantly reorganized French sub-regional codes. I use the authoritative Eurostat GISCO GeoJSON file (NUTS_RG_01M_2024_4326_LEVL_3) to construct the mapping from NUTS-3 codes to French département codes. Key changes from the previous classification include:

- FRB now designates Centre-Val de Loire (previously associated with different regions)
- FRF sub-regions were renumbered: FRF1 = Alsace, FRF2 = Champagne-Ardenne, FRF3 = Lorraine
- FRI sub-regions were renumbered: FRI1 = Aquitaine, FRI3 = Poitou-Charentes
- FRK (Auvergne-Rhône-Alpes) was completely reorganized
- FRM now designates Corsica

The mapping covers all 96 metropolitan départements with a perfect one-to-one correspondence to NUTS-3 codes.

A.2 Election Data

Commune-level first-round results are obtained from `data.gouv.fr` in Parquet format. I identify Rassemblement National/FN candidates using a combination of party codes (nuance codes: FN, RN, EXD, LFN, LRN, LEPE) and candidate name matching (Le Pen, Bardella, and list-specific keywords).

Sample construction. Starting from the universe of commune × election observations, I apply the following filters:

1. Restrict to metropolitan France (96 départements, excluding overseas territories)
2. Require positive registered voters and positive valid votes
3. Require RN vote share to be non-missing (communes where the RN did not field a candidate are dropped)

With approximately 37,000 communes across 6 elections, the theoretical maximum is roughly 222,000 observations. The actual panel contains 214,966 observations because some communes did not have an RN candidate in every election and others had incomplete returns. After matching to département-level exposure variables, the regression sample contains 212,803 observations.

A.3 Fuel Vulnerability

I construct département-level fuel vulnerability using INSEE data on commuting-related CO_2 emissions. The variable measures the average metric tons of CO_2 equivalent emitted per worker per year for home-to-work commuting, based on the dominant transport mode and commute distance.

To convert CO_2 emissions into a monetary burden, I use the legislated carbon tax schedule:

A.4 Filosofi Income Data

Département-level median standard of living (*niveau de vie médian*) comes from the INSEE Filosofi database (2021 vintage). This is defined as household disposable income per consumption unit (OECD-modified equivalence scale). I merge this at the département level and interact with the post-carbon-tax indicator for the income-control robustness specification.

Table 5: France Carbon Tax Schedule, 2013–2024

Year	Rate (€/tCO_2)	TICPE Diesel (c/L)	TICPE SP95 (c/L)	Carbon Component Diesel (c/L)
2013	0.0	42.84	60.69	0.00
2014	7.0	42.84	60.69	1.86
2015	14.5	46.82	62.41	3.84
2016	22.0	49.81	64.12	5.83
2017	30.5	53.07	65.07	8.09
2018	44.6	59.40	68.29	11.82
2019	*44.6*	*59.40*	*68.29*	*11.82*
2020	*44.6*	*59.40*	*68.29*	*11.82*
2021	*44.6*	*59.40*	*68.29*	*11.82*
2022	*44.6*	*59.40*	*68.29*	*11.82*
2023	*44.6*	*59.40*	*68.29*	*11.82*
2024	*44.6*	*59.40*	*68.29*	*11.82*

Notes:
Carbon tax rates from the Contribution Climat-'Energie (CCE), enacted in the 2014 Finance Law. TICPE = Taxe Int'erieure de Consommation sur les Produits 'Energ'etiques. Rates were frozen at the 2018 level following the textitGilets Jaunes protests (italicized rows). Carbon component diesel computed as rate *times* 2.651 kg CO_2/L. Sources: DGEC; Journal Officiel.

A.5 Variable Definitions

Table 6: Variable Definitions

Variable	Definition
RN vote share	First-round votes for FN/RN candidate or list, divided by total valid votes, expressed as a percentage
Turnout	Total votes cast divided by registered voters, expressed as a percentage
Own fuel exposure	Standardized commuting CO_2 emissions (tCO_2e per worker per year) at the département level
Network fuel exposure	SCI-weighted average of connected départements' commuting CO_2, standardized
Post carbon tax	Indicator equal to 1 for elections in 2017 and later (carbon rate $\geq$ €30.50/tCO_2)
Post *Gilets Jaunes*	Indicator equal to 1 for elections in 2019 and later (after the *Gilets Jaunes* crisis)

Notes: Both exposure variables are standardized to mean zero, unit variance using the full commune-level panel. Network exposure is constructed per Equation 6.

B. Identification Appendix

B.1 SCI Network Structure

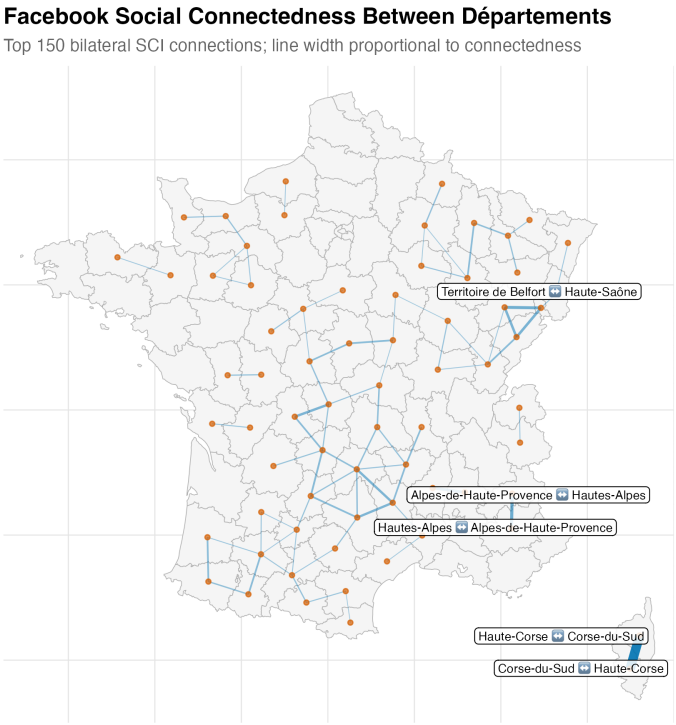

Figure 4: Top SCI Connections Between French départements
Notes: Lines represent the strongest 50 SCI connections between metropolitan départements. Line thickness is proportional to SCI score. The network reveals a mix of geographic proximity (neighboring départements) and long-distance connections (e.g., Paris to provincial capitals).

Figure 4 visualizes the strongest SCI connections. The network has a hub-and-spoke structure with Paris at the center, but also includes important regional connections (e.g., between Alsatian départements, between the Basque Country and neighboring départements).

The table above lists the 20 strongest SCI connections. The highest-SCI pairs are typically neighboring small or rural départements within the same region (e.g., Corse-du-Sud – Haute-Corse, Alpes-de-Haute-Provence – Hautes-Alpes, Haute-Saône – Territoire de Belfort). These pairs have few Facebook users, so any cross-département friendship generates a high relative probability. Large urban départements (Paris, Lyon) have lower per-pair SCI because their populations are spread across many connections.

Table 7: Top 20 Département Pairs by Facebook Social Connectedness Index

From	To	SCI Score
Corse-du-Sud (2A)	Haute-Corse (2B)	5,642,059
Alpes-de-Haute-Provence (04)	Hautes-Alpes (05)	1,875,055
Haute-Saône (70)	Territoire de Belfort (90)	1,758,049
Ardèche (07)	Drôme (26)	1,575,958
Doubs (25)	Territoire de Belfort (90)	1,515,361
Doubs (25)	Haute-Saône (70)	1,458,021
Creuse (23)	Haute-Vienne (87)	1,422,104
Cantal (15)	Lozère (48)	1,381,722
Aveyron (12)	Lozère (48)	1,358,084
Corrèze (19)	Lot (46)	1,081,600
Aveyron (12)	Lot (46)	1,037,831
Cher (18)	Nièvre (58)	1,035,181
Doubs (25)	Jura (39)	1,027,837
Landes (40)	Pyrénées-Atlantiques (64)	1,023,230
Gers (32)	Hautes-Pyrénées (65)	1,015,254
Haute-Loire (43)	Lozère (48)	999,693
Corrèze (19)	Haute-Vienne (87)	997,608
Haute-Marne (52)	Meuse (55)	982,623
Loire (42)	Haute-Loire (43)	952,966
Cher (18)	Indre (36)	951,885

Notes:
Social Connectedness Index (SCI) from Bailey et al. (2018), measured at the NUTS-3 level (French d'epartements). SCI captures the relative probability of a Facebook friendship between users in each pair of regions. Higher values indicate stronger social ties. Source: Facebook Data for Good, NUTS 2024 classification.

B.2 Shift-Share Diagnostics

Following Goldsmith-Pinkham et al. (2020), I decompose the shift-share instrument into its constituent "shares" (SCI weights) and assess their exogeneity to baseline outcomes. Table 8 reports the full set of diagnostics.

Table 8: Bartik/Shift-Share Diagnostics

Diagnostic	Value
Panel A: Rotemberg Weight Concentration	
Top-5 Rotemberg weight sum	0.265
Herfindahl-Hirschman Index (HHI)	0.025
Effective number of sources (1/HHI)	39.4
Panel B: Share Exogeneity	
F-statistic (top-5 shares → 2012 RN vote)	2.49
p-value	0.054

Notes: Panel A reports Rotemberg weight concentration following Goldsmith-Pinkham et al. (2020). The effective number of sources measures instrument diversification; values well above 1 indicate that no single source département dominates. Panel B tests whether the SCI shares of the top-5 Rotemberg-weight source départements predict baseline (2012) Rassemblement National vote share. Failure to reject at 5% supports approximate share exogeneity.

Rotemberg weights. The Rotemberg weight for source département j measures its contribution to the overall network exposure estimate. In a well-diversified instrument, no single source should dominate. The effective number of sources (39) is well above the threshold of concern, though the top-5 concentration (26.5%) indicates that some départements contribute disproportionately.

Share exogeneity. I test whether the top-5 Rotemberg-weight source départements' SCI shares predict baseline (2012) Rassemblement National vote share. The F-statistic is 2.49 ($p = 0.054$). The failure to reject at 5% provides evidence that the SCI shares are approximately exogenous, though the marginal significance at 10% warrants caution. The correlation between baseline Rassemblement National support and SCI shares is modest, suggesting that the network structure is not strongly selected on pre-existing political preferences.

B.3 Pre-Trends

The event study in Figure 1 provides the primary pre-trends diagnostic. The 2012 coefficient for network exposure (relative to 2014) is large and negative (-1.45, $p < 0.01$), indicating that the association between network fuel exposure and Rassemblement National support was significantly weaker in 2012 than in 2014. This is consistent with the carbon tax activating the network channel, but with only one pre-treatment period, a pre-existing trend cannot be

ruled out. I interpret this conservatively: the evidence supports a shift in the network–voting relationship around the time of the carbon tax introduction, but the causal attribution rests on the assumption that no other confounding factor generated the same shift.

B.4 Parallel Trends Assessment

The parallel trends assumption requires that départements with different levels of network fuel exposure would have experienced similar Rassemblement National vote share trajectories in the absence of the carbon tax. Several pieces of evidence support this:

1. The 2012 event study coefficient is large and negative, indicating a shift in the network–voting relationship between 2012 and 2014. While this is consistent with the carbon tax activation, a pre-existing trend cannot be ruled out with a single pre-treatment observation.

2. The placebo (turnout) shows no effect, suggesting that the SCI is not correlated with general electoral trends.

3. The income-controlled specification produces similar results, ruling out the concern that SCI-connected départements share income-driven political trends.

4. The leave-one-out exercise shows no influential outliers.

Together, these results support the identifying assumption but cannot definitively establish it. The strongest evidence comes from the placebo test (network exposure predicts Rassemblement National support but not turnout) and the distance restriction (the effect survives when local connections are excluded). The event study pattern is suggestive but not dispositive given the single pre-treatment observation.

C. Robustness Appendix

C.1 Alternative SCI Normalization

In the baseline specification, I use row-normalized SCI weights ($w_{dj} = \text{SCI}_{dj} / \sum_k \text{SCI}_{dk}$). An alternative is to use raw SCI scores, which preserve information about the *absolute* level of social connectedness. Départements with higher total SCI (i.e., more socially connected overall) would receive higher network exposure under this normalization.

The correlation between normalized and raw SCI-weighted fuel exposure is 1.00, indicating that the two measures are effectively identical. This occurs because the raw SCI scores

are already proportional to the normalized weights within each origin département. The regression results using raw SCI weights are correspondingly identical to the baseline (own: 0.17, network: 0.48***).

C.2 Binscatter: Network Exposure and RN Change

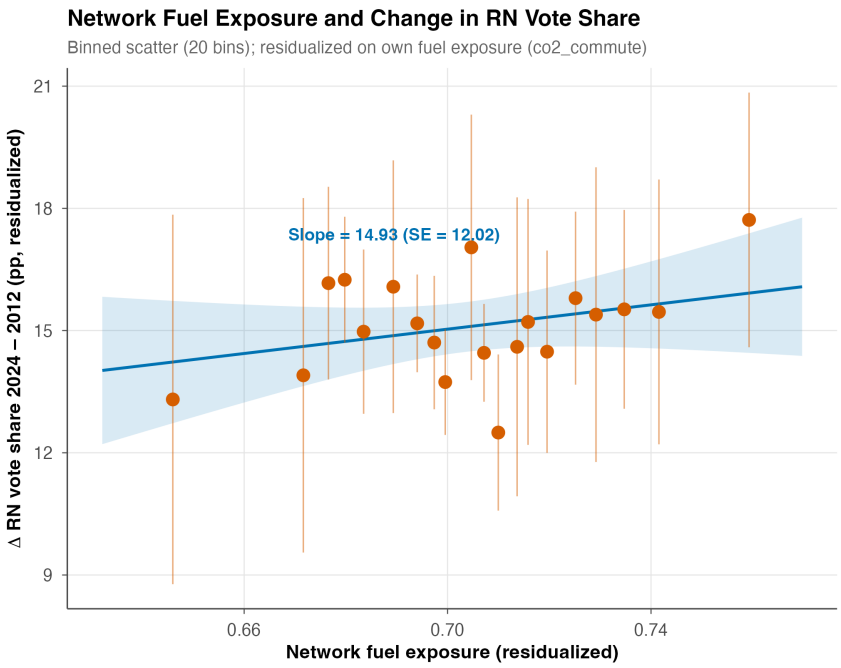

Figure 5: Binscatter: Network Fuel Exposure and Change in Rassemblement National Vote Share *Notes:* Each point represents a ventile (5%) bin of département-level network fuel exposure (SCI-weighted commuting CO_2). The y-axis shows the change in Rassemblement National vote share from 2012 to 2024 (percentage points), residualized with respect to département and election fixed effects. The line shows a linear fit. The positive relationship is consistent with the regression results: départements with higher network fuel exposure experienced larger Rassemblement National gains.

Figure 5 provides a non-parametric visualization of the relationship between network fuel exposure and Rassemblement National vote share changes. The positive slope is visible and approximately linear, consistent with the parametric specification.

C.3 Leave-One-Out Distribution

The leave-one-out exercise drops each département in turn and re-estimates the main specification. The distribution of the 96 network coefficients is tightly clustered:

- Mean: 0.478 (virtually identical to full sample)

- Standard deviation: 0.028
- Minimum: 0.420 (dropping département with highest leverage)
- Maximum: 0.556
- 100% statistically significant at the 5% level

This confirms that the result is not driven by any single département. The narrow range (0.42–0.56) implies that the estimate is highly stable.

C.4 Post-*Gilets Jaunes* Specification

The post-*Gilets Jaunes* specification uses an indicator equal to one for elections in 2019 and later. This captures the period after the *Gilets Jaunes* crisis made the carbon tax politically salient. The network coefficient (0.38, $p < 0.05$) is somewhat smaller than the baseline (0.48), which includes the 2017 election. This suggests that the network effect was already present before the *Gilets Jaunes* crisis—the tax itself was generating backlash through social networks, and the protests amplified rather than created this dynamic.

D. Structural Model Details

D.1 SAR Model Estimation

The spatial autoregressive (SAR) model is estimated by maximum likelihood following Lee (2004) and LeSage and Pace (2009). The key computational step is evaluation of the log-determinant $\log|I - \rho W|$.

I compute this using the eigendecomposition of W. Let $\lambda_1, \ldots, \lambda_N$ be the eigenvalues of W. Then:

$$\log|I - \rho W| = \sum_{i=1}^{N} \log(1 - \rho\lambda_i) \tag{11}$$

The row-normalized SCI weight matrix is not symmetric in general (since $\mathrm{SCI}_{dj}/\sum_k \mathrm{SCI}_{dk} \neq \mathrm{SCI}_{jd}/\sum_k \mathrm{SCI}_{jk}$), so its eigenvalues may be complex. The relevant properties are:

- The spectral radius $\varrho(W) = 1$ (since W is row-stochastic, 1 is an eigenvalue; all others have modulus ≤ 1)
- For $\rho < 1/\varrho(W) = 1$, the matrix $(I - \rho W)$ is invertible
- The log-determinant $\log|I - \rho W| = \mathrm{Re}\left[\sum_i \log(1 - \rho\lambda_i)\right]$, computed by pairing complex conjugate eigenvalues and taking real parts

The estimation proceeds by concentrating the log-likelihood with respect to β and σ^2, yielding a one-dimensional optimization over ρ:

$$\hat{\rho} = \arg\max_{\rho \in [0,1)} \left\{ T \sum_{i=1}^{N} \log(1 - \rho\lambda_i) - \frac{NT}{2} \log \hat{\sigma}^2(\rho) \right\} \tag{12}$$

where $\hat{\sigma}^2(\rho)$ and $\hat{\beta}(\rho)$ are the OLS estimates from regressing $(\tilde{s} - \rho W \tilde{s})$ on $\tilde{X}$ for each candidate ρ.

I use a two-stage approach: a coarse grid search over $\rho \in \{0, 0.01, 0.02, \ldots, 0.99\}$ followed by Brent optimization in a neighborhood of the grid maximum. The standard error of $\hat{\rho}$ is computed from the numerical Hessian of the concentrated log-likelihood.

D.2 Cross-Sectional Variation in ρ

To assess whether network dependence varies across elections, I estimate the SAR model separately for each of the five post-introduction election cross-sections (2014–2024). This provides election-specific estimates of ρ.

The cross-sectional estimates reveal high spatial dependence across all elections, with ρ ranging from approximately 0.95 to 0.98. The small variation suggests that the SCI-based network structure consistently captures spatial correlation in voting patterns, regardless of election type or the specific political context.

D.3 Impulse Response Analysis

The spatial multiplier matrix $(I - \hat{\rho}W)^{-1}$ encodes the full structure of network amplification. Following the impact decomposition of LeSage and Pace (2009), I report three measures: the *direct effect* (average diagonal element of $(I - \hat{\rho}W)^{-1}$, capturing own feedback), the *total effect* (average row sum, capturing the full network-amplified impact), and the *indirect effect* (total minus direct, capturing spillovers to other départements).

For row-stochastic W, the total effect multiplier equals $1/(1-\hat{\rho}) \approx 33.3$. This represents the sum of all impacts—direct and indirect—that a unit shock to one département's fundamentals generates across the entire network. The direct effect (average diagonal) is smaller because $w_{dd} = 0$ prevents first-order self-feedback; own effects arise only through higher-order network loops. The cross-multiplier structure reveals that shocks to well-connected départements (high SCI out-degree) generate larger total effects, consistent with the hub-and-spoke structure of the French social network.

D.4 Model Fit

The SAR model substantially outperforms OLS ($\rho = 0$):

- OLS log-likelihood: $-1{,}572.3$
- SAR log-likelihood: -989.2
- Improvement: 583.1 log-likelihood units
- LR test: $\chi^2 = 1{,}166.1$ ($p < 0.001$)

The dramatic improvement in fit confirms that the SCI-based network structure captures an important source of cross-département correlation in Rassemblement National vote shares that OLS cannot account for.

E. Additional Figures and Tables

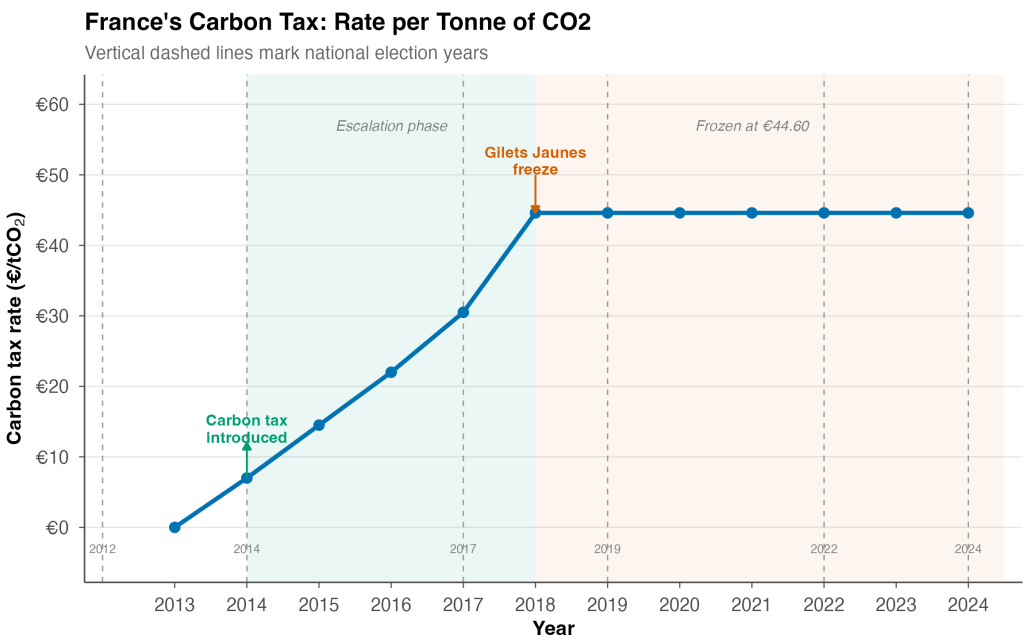

Figure 6: France Carbon Tax Rate, 2013–2024

Notes: Carbon tax rate (Contribution Climat-Énergie) in euros per ton of CO_2, 2013–2024. The rate rose from €7 in 2014 to €44.60 in 2018, then was frozen following the *Gilets Jaunes* crisis. The dashed line shows the originally planned trajectory.